

BRAC UNIVERSITY
Department of Computer Science and Engineering

Quiz 02
Semester: Fall 2024

Time: 20min
Full Marks:15

Name:	ID:	Section:
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CSE 251: Electronic Devices and Circuits

Q1. [CO2] A voltage waveform $V_i = 15\sin(200\pi t)$ V is fed into a full-wave rectifier with a load resistance $R = 5 \text{ k}\Omega$. Silicon diodes are used in this circuit for which the forward drop is $V_{DO} = 1 \text{ V}$.

- a. **Draw** the rectifier circuit and the voltage transfer characteristics of the rectifier. [5]
- b. **Illustrate** the input and output waveforms in separate graphs. **Label** the graph and indicate the voltage levels properly. [5]
- c. **Calculate** the DC/Average value of the output. [5]
- d. A capacitor is now added to reduce the fluctuation of the output voltage, which makes the peak to peak ripple voltage 4% of the maximum output voltage V_P . **Deduce** the value of the capacitor from the given data. [5]